Contact

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Top Skills

Forward Deployed Engineering
Retrieval-Augmented Generation (RAG)
LangGraph

Certifications

Inferential Statistics
Machine Learning
Software as a Service - Cloud Computing
Microsoft Certified: Azure Fundamentals

Honors-Awards

Got NSF presenter award for presenting a research paper on Alzheimer's clinical trial in the Machine Learning Science and Engineering conference co-hosted by CMU and Georgia Tech.

Gained recognition from senior management for exceptional work with the Spot Award in TCS.

Publications

Maker, M.C. Prakash, J., Wright, E. Jordan, K., May, T., Raja, M., Mitchell, C.S., ‘Quantifying amyloid beta reduction necessary in transgenic Alzheimer’s mice using BACE inhibitors.’ Journal of Alzheimer’s Disease.

Prakash, J., Mitchell, C.S., ‘Phenotyping patient characteristics in mild cognitive decline, Alzheimer’s, and normative aging.’

Biomedical Text Link Prediction for Drug Discovery: A Case Study with COVID-19

Jayant Prakash

AI Engineer | Forward Deployed Engineer (FDE) | Generative AI & Agentic AI | LLMs, RAG, AI Agents | LangGraph, Python | AWS, Azure

Atlanta, Georgia, United States

Summary

I am a Forward Deployed Engineer (FDE) and AI Engineer with experience across software engineering, machine learning, and production AI systems. I specialize in translating complex business and operational challenges into scalable AI solutions from technical discovery and rapid prototyping to architecture, enterprise integration, evaluation, deployment, and continuous improvement.My core focus includes Generative AI, Agentic AI, AI Agents, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and production machine learning. I build AI applications that integrate intelligent workflows with enterprise APIs, backend services, retrieval systems, tool calling, human-in-the-loop controls, and production infrastructure.Most recently, I architected and developed an agentic LLM-powered outage intelligence and operational automation solution using Python, AWS, Open AI models, and LangGraph. The system reduced manual triage by approximately 70%, improved response time by 45%, and achieved approximately 90% classification accuracy.My experience includes multi-step agentic workflows, RAG pipelines, AI tools, REST APIs, FastAPI backend services, AI evaluation, LLM-as-judge, prompt regression testing, hallucination reduction, guardrails, and Responsible AI practices.At Comcast, I developed Generative AI and machine learning solutions for enterprise automation, including an LLM-powered dispute-resolution system that reduced processing time by approximately 70% and automated approximately 35% of disputes, along with recommendation systems, backend APIs, ML workflows, CI/CD, and production monitoring.My technical experience includes Python, Java, LangGraph, LangChain, RAG, Pinecone, FAISS, Open AI, Llama, Claude, Gemini, FastAPI, REST APIs, Docker, AWS, Azure, GitHub Actions, PyTorch, and scikit-learn.I enjoy working at the intersection of AI engineering, business needs, product thinking, and hands-on software development—understanding real-world problems and turning successful prototypes

into reliable production AI systems with measurable impact.I hold an MS in Computer Science from Georgia Tech with a specialization in AI and welcome connections with engineering leaders, hiring managers, founders, recruiters, and AI professionals working in AI Engineering, Forward Deployed Engineering, Applied AI, Generative AI, and Agentic AI.

Experience

AT&T

Forward Deployed Engineer/AI Engineer
July 2025 - Present (1 year 3 months)
Georgia, United States

• Architected, developed, and deployed a scalable agentic LLM-powered outage intelligence and operational automation system using Python, AWS, OpenAI models, and LangGraph, reducing manual triage by approximately 70% and improving response time by approximately 45%.

• Collaborated with business, operations, and engineering teams to identify AI automation opportunities, translate operational challenges into technical solutions, and integrate AI services into enterprise production systems.

• Architected agentic AI workflows using Python, LangGraph, LangChain, RAG, and Pinecone for multi-step orchestration, retrieval, tool integration, and validation across telecom operational use cases.

• Developed REST API integrations connecting AI agents with operational and incident data and ServiceNow workflows, enabling automated incident enrichment, orchestration, and response management.

• Engineered production LLM workflows with structured outputs, tool calling, failure handling, validation, prompt engineering, and Responsible AI guardrails, achieving approximately 90% classification accuracy.

• Built custom AI tools for parsing, enrichment, workflow automation, and operational execution, increasing automation coverage from approximately 40% to more than 75%.

• Implemented human-in-the-loop approval and validation workflows to support secure and reliable AI execution in enterprise production environments.

• Established AI evaluation and testing frameworks using unit, integration, prompt regression, and LLM-as-judge techniques, reducing production failures by approximately 60%.

• Developed scalable FastAPI-based backend AI services, containerized with Docker and deployed on AWS to support enterprise integrations, production rollout, monitoring, logging, and reliability validation.

• Implemented GitHub Actions CI/CD pipelines supporting automated testing, containerized deployment, prompt regression validation, and continuous delivery of AI/LLM applications.

Comcast

AI Engineer/Machine Learning Engineer
March 2021 - March 2025 (4 years 1 month)
Atlanta, GA

• Designed and developed a Generative AI-driven dispute-resolution and operational automation solution, reducing processing time by approximately 70% and delivering operational efficiencies through automation.

• Built scalable Python-based LLM pipelines using Llama 70B and Llama 3.1 models to process customer communications and automate operational workflows, enabling approximately 35% of disputes to be handled through automation.

• Engineered a recommendation system and AI-enabled workflow automation solution that improved technician acceptance from 23% to 95%, reducing manual allocation effort and improving operational outcomes.

• Developed Python REST APIs and backend AI services to ingest, preprocess, and deliver scalable machine learning and enterprise AI workflows.

• Developed deep-learning-based sentiment classification models using embeddings and neural networks in PyTorch, achieving 98% test accuracy following model optimization.

• Developed SQL queries using Teradata to extract and prepare enterprise datasets for machine learning development.

• Implemented MLOps and CI/CD workflows using GitHub Actions, with Prometheus and Grafana supporting production monitoring, performance visualization, reliability, and continuous improvement.

Georgia Institute of Technology
Machine Learning Engineer/Data Scientist
January 2019 - May 2020 (1 year 5 months)
Atlanta

• Developed machine learning models to predict Alzheimer’s cognitive decline with an 85% test R2 using XGBoost in Python.
• Utilized PCA and k-means to segment Alzheimer’s patients into four clusters, enabling personalized treatment.
• Deployed machine learning model into production using AWS Sagemaker for real-time patient care.

Tata Steel
Machine Learning Engineer/Data Scientist
August 2017 - July 2018 (1 year)

• Developed convolution neural network in TensorFlow to classify surface defects on steel images, improving automation and maintaining high quality in production.
• Tuned hyperparameters using Keras-tuner, achieving a test accuracy of 88%.

American Airlines
Machine Learning Engineer/Data Scientist
July 2015 - August 2017 (2 years 2 months)

• Implemented k-means clustering in Python on airline customer data to enhance targeted marketing campaigns.
• Drove a 10% increase in customer retention through data-driven market segmentation.

SABMiller
Software Engineer
November 2012 - June 2015 (2 years 8 months)

• Developed SAP front-end application ‘Purchase Order Approval System using Java, enabling users to view, approve, and reject queued orders pending approval.

• Created the ‘Waste Calculation System’ ALV report using ABAP to calculate internal process waste; and reduced wastage by 4% over 12 months.

Education

Georgia Institute of Technology

Master of Science - MS, Computer Science(Artificial Intelligence) · (2018 - 2020)

SRM University

Bachelor of Technology (B.Tech.), Computer Science · (2008 - 2012)